# HOTELREVAI
# AI DRIVEN REVENUE ANALYSIS FOR HOTELS

# PRESENTED BY (TEAM B):

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## Project Overview & Objectives

**Project Overview :**

This analytics project focuses on analyzing hotel booking data to discover meaningful business insights. The system studies customer booking behavior, revenue trends, market segment performance, and seasonal demand patterns. The goal is to transform raw data into **useful insights that help hotel managers make better decisions.**

**Key Objectives •**

- Transform raw CSV hotel data into structured insights
- Build an efficient data model with proper relationships
- Create important business KPIs using DAX
- Develop an interactive Power BI dashboard for decision making

# Milestone 1:

**Orientation & Data Loading** :

**Orientation Phase**

- Understood the project objectives and scope
- Identified key business problems in hotel revenue analysis
- Explored dataset structure and schema

**Dataset Information**

The project uses a Hotel Booking Dataset (CSV format) that includes:

- Booking and arrival dates
- Market segment classification
- Customer types and preferences
- Average Daily Rate (ADR)

**Data Loading Process**

The dataset was imported into Power BI Desktop for analysis.

Steps involved:

- Importing the CSV dataset
- Checking and correcting column data types
- Verifying data completeness and quality
- Preparing the dataset for further analysis

This created a **baseline dataset  ready for preprocessing**.

## Data Cleaning & Preprocessing

To ensure accurate analysis, the dataset was cleaned and prepared.

**Steps Performed**

### 1. Remove Missing Values

Null or incomplete records were removed.

### 2. Correct Data Types

Incorrect formats were corrected.

### 3. Remove Duplicates

Duplicate records were eliminated.

### 4. Standardize Names

Column names and values were normalized.

### 5. Handle Invalid Data

Incorrect categorical values and outliers were corrected.

### 6. Create Calculated Fields

New fields were created for deeper analysis.

# Milestone 2:

**Data Modeling Feature Selection**

**Important features selected for analysis include:**

- Booking Date and Arrival Date
- Room Type classification
- Guest count and composition
- Average Daily Rate (ADR)
- Lead Time analysis
- Cancellation status
- Market segment category
- Customer type segmentation These features provide valuable insights into hotel performance.

## Structured Data Model

A structured data model was created to improve analysis.

**Key Steps**

- Created relationships between tables
- Built year-wise and month-wise structures
- Developed a comprehensive date hierarchy
- Ensured optimized relationships between data tables This model supports efficient.

## Dashboard Development

An interactive Power BI dashboard was developed.

**Core KPIs**

- Total Bookings
- Average Stay Duration
- Average Lead Time
- Average Daily Rate (ADR)

**Visual Analysis**

- Bookings by Market Segment
- Bookings by Month
- Bookings by Customer Type
- Revenue Trends

## Interactive Dashboard Features

The dashboard provides interactive analysis through:

- Year slicer (2015–2017)
- Dynamic filtering options
- Cross-visual interactions
- Clean and user-friendly interface .

These features allow hotel managers to easily explore data insights.

## Milestone 3:

**DAX & Business Metrics**

**What is DAX?**

DAX (Data Analysis Expressions) is the formula language used in Power BI

It allows users to:

- Create calculated measures
- Perform complex calculations
- Generate business KPIs

DAX enables advanced data analysis and reporting.

**DAX Measures Created**

Several important measures were created:

- **Total Bookings:**
  Total number of confirmed hotel reservations.
- **Total Revenue:**
  Total revenue generated from all hotel bookings.
- **Average Stay:**
  Average number of days guests stay at the hotel.
- **Average Lead Time:**
  Average number of days between the booking date and the arrival date.
- **Average Daily Rate (ADR):**
  Average room price charged per night.
- **Cancellation Rate:**
  Percentage of bookings that were cancelled.

**Business Metrics Generated**

The project generated several valuable insights.

**Revenue Trends:**

Monthly and yearly revenue patterns help forecast future revenue.

**Seasonal Demand:**

Identifies peak and off-season booking periods.

**Customer Segments :**

Analyzes performance by market segment and customer type.

**Channel Contribution:**

Examines which booking channels generate the most reservations

**Key Insights & Business Impact Key Insights**

**Top Channel:**

Online Travel Agents generate the highest bookings.

**Customer Profile:**

Transient customers dominate the market

**Peak Season:** Highest demand occurs in mid-year months

**Average Stay:** Most guests stay about 3 days

**Booking Window:** Customers usually book around 105 days.

**Business Impact**

The project provides several benefits:

- Revenue optimization through better pricing strategies
- Improved forecasting for demand planning
- Data-driven pricing based on seasonal trends
- Improved operational efficiency Better understanding of customer behavior.

**Project Outcomes & Value Delivered**

**Key Achievements**

3 Milestones Completed From data loading to dashboard development.

**5+ KPIs Created**

Important business metrics for monitoring performance.

**3 Years of Data Analyzed**

Historical dataset from 2015–2017.

**20+ Visualizations Built**

Interactive charts and graphs.

## Milestone 4:

**AI Insights & Visualization**

This milestone focuses on transforming hotel booking data into strategic visual intelligence.

**Key Components**

- Interactive data visualizations
- AI-powered analysis
- Dynamic dashboards
- Business performance insights

These features help hotel managers quickly understand performance

**Dashboard Components**

**KPI Cards**

- Total Revenue
- Average Revenue
- Total Bookings

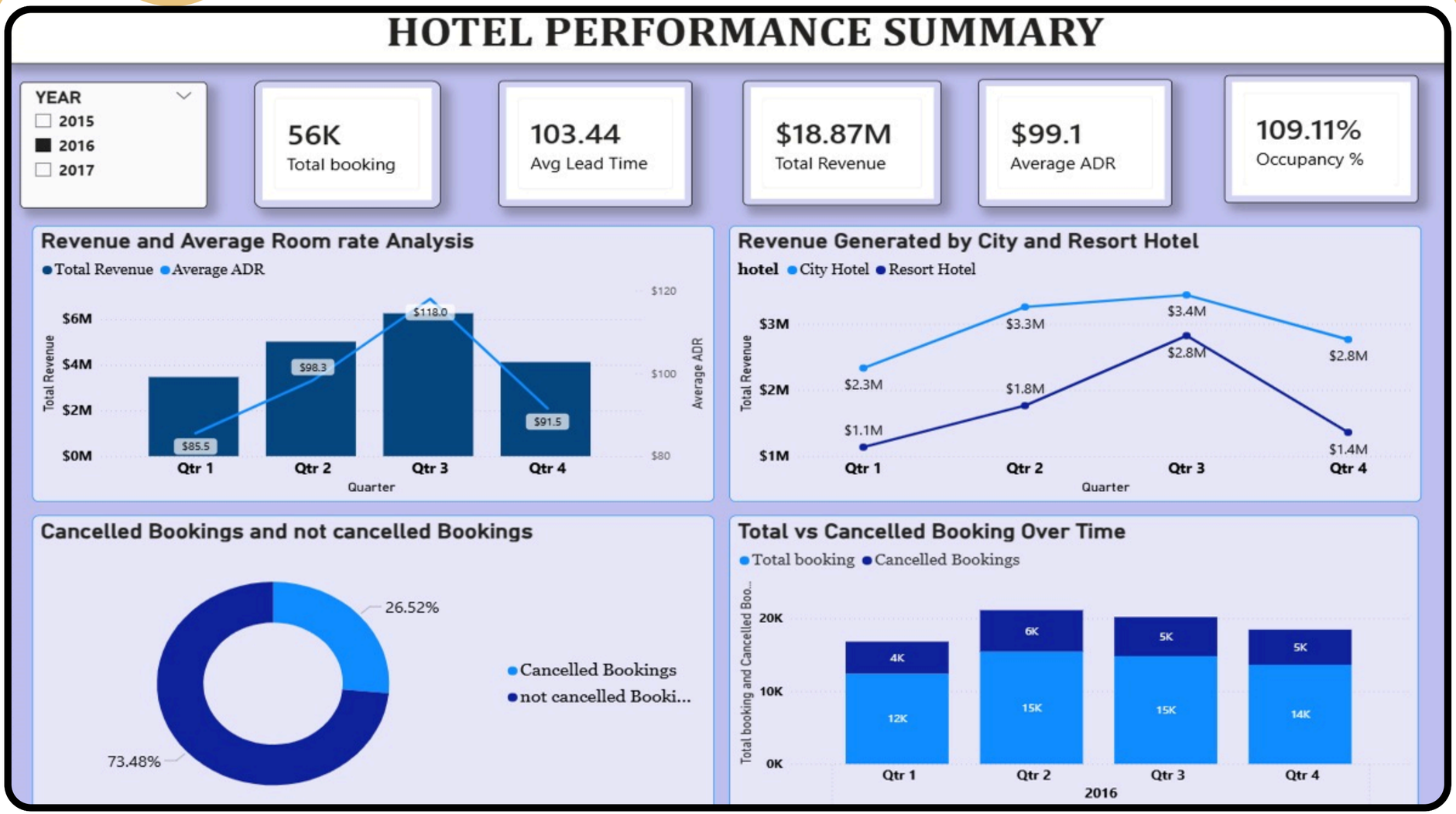
**Charts & Trends**

- Revenue by Year
- Bookings by Hotel Type
- Customer Distribution

**Interactive Filters**

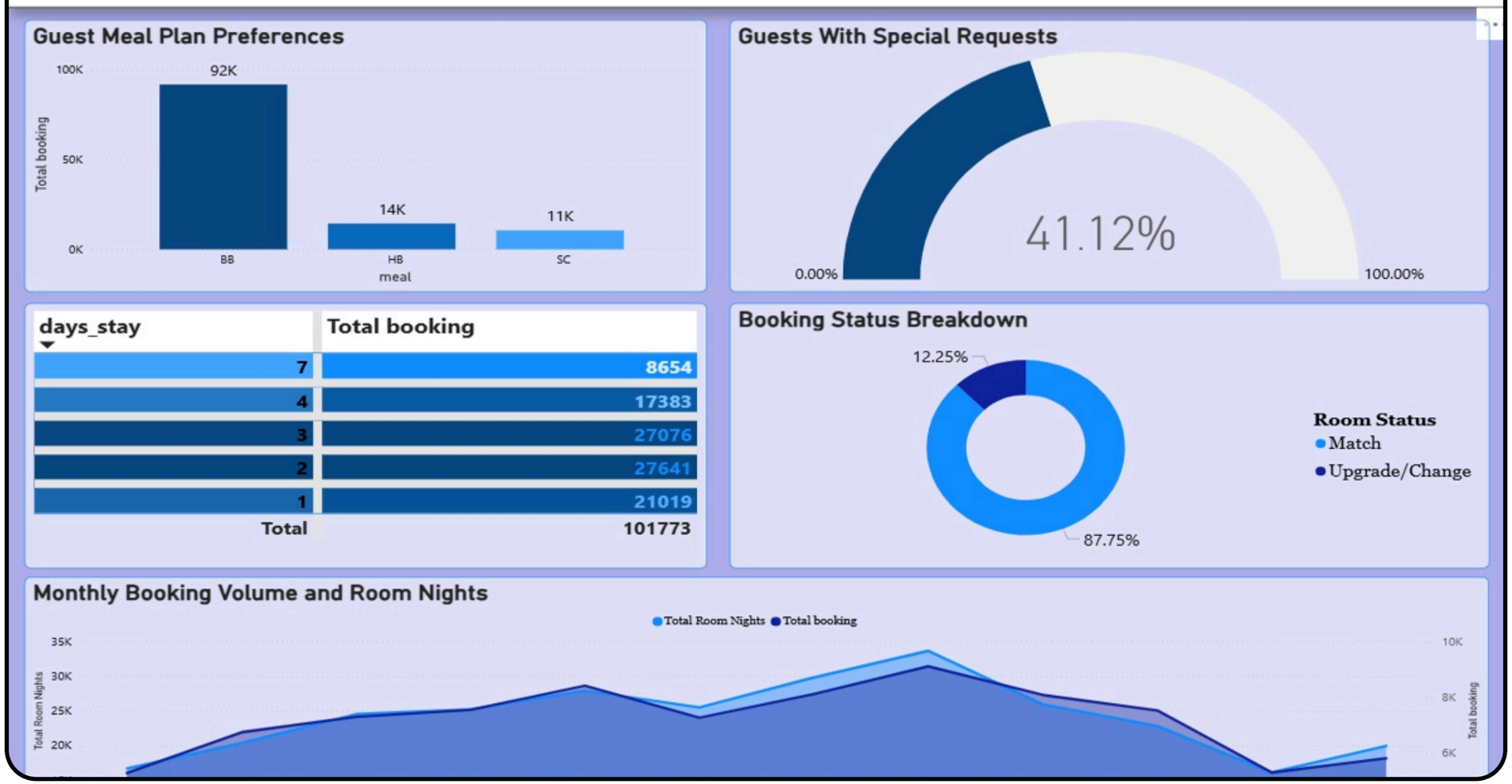
- Year selection
- Hotel type filtering
- Customer segmentation

# HOTEL PERFORMANCE SUMMARY

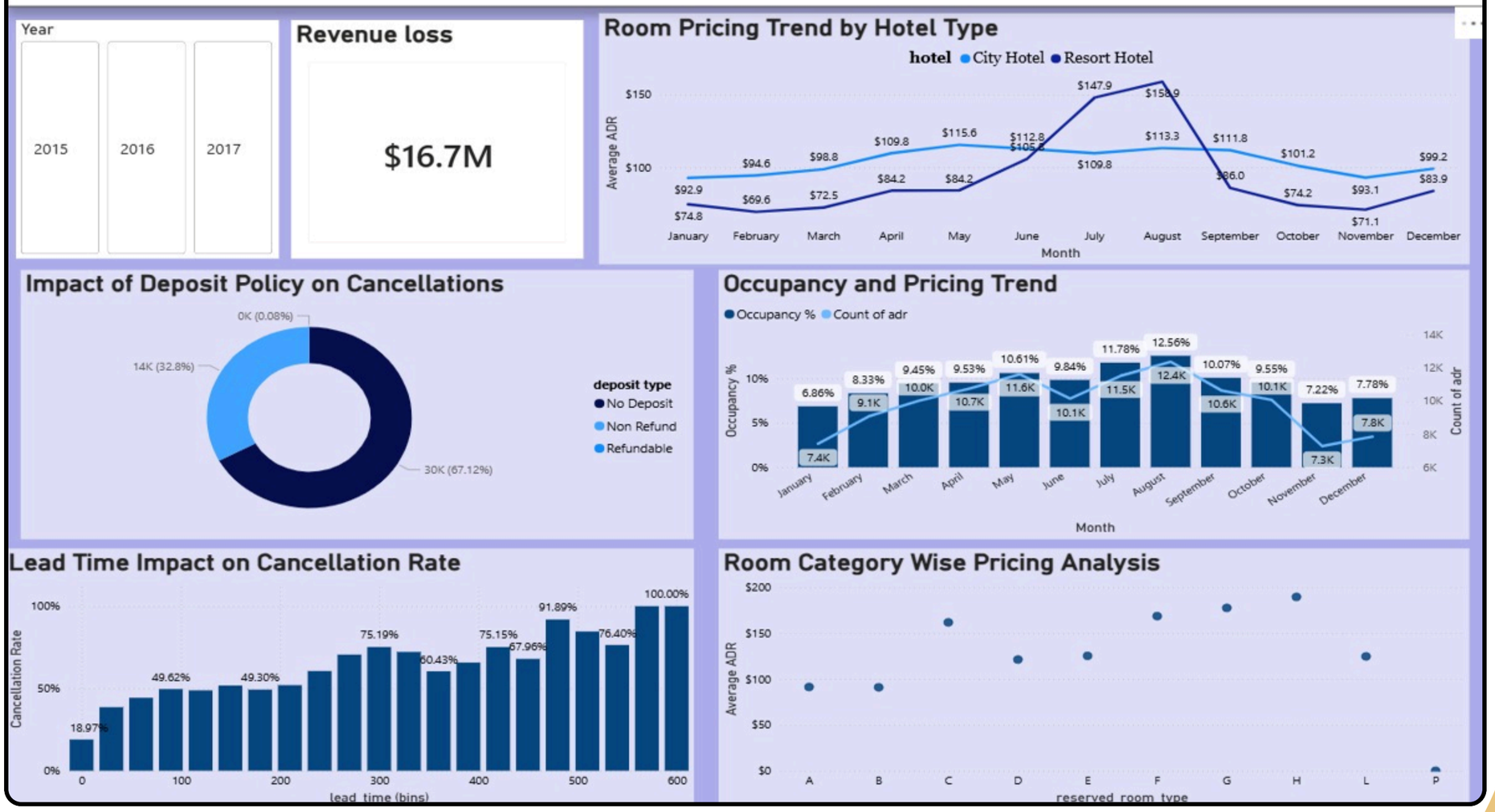


# Guest Profiling & Market Segmentation

# Booking Behavior & Stay Preferences



# Revenue & Cancellation



**Driving Data-Driven Decisions**

The system converts raw data into actionable insights.

P**rocess Flow: Raw Data → AI Processing → Visual Insights → Strategic Decisions**

This helps hotel management move from reactive decision-making to proactive strategy.

**Conclusion**

HotelRevAI demonstrates how **AI, data analytics, and visualization** can transform raw hotel booking data into valuable business insights.

By using Power BI dashboards and advanced analytics, hotels can:

- Improve revenue management
- Optimize pricing strategies
- Understand customer behavior
- Make better data-driven decisions

# Thank You